



Focus And Zoom

The focal length of a lens is the distance between the lens and the CCD. Using an infra-red detector, the distance to the subject being shot is determined and a series of motors automatically change the lens' focus to accommodate a sharp picture. A manual over-ride is of course available.

By increasing the focal length of the lens, a scene can be magnified. This is known as the optical zoom of a lens; apart from this some camcorders also have a digital zoom. This zoom takes place on the CCD of a camera: the number of pixels used for the area to be magnified is increased, thus creating a larger image. This causes a loss in information, and a digital zoom is thus not as sharp.

this manner is then fed to an on-board computer which interpolates—guesses the final value based on already gathered information—the available colour information to form the final image. The interpolation invariably loses out on information, and thus you get what you paid for in terms of image quality.

Now that we have translated the real world into a series of electrical signals, a layer sitting below the CCD takes up the charges and writes them onto the recording medium. A digital

Analogue recording fades with time, while a digital recording doesn't degrade at all with the passage of time.

camcorder employs an analogue-to-digital (ADC) converter at this stage, which converts the analogue electrical charges into bits of data (0s and 1s) and then stores them digitally.

In terms of operation this is the only thing that separates a digital camcorder from an analogue one. Practically, the differences are several. Most importantly, an analogue recording fades with time while a digital one stays the same with age. A digital record of memories also affords greater flexibility: take it to a digital tool such as Adobe Premiere or VirtualDub to add effects, transitions, time-shifts, etc; e-mail it, record it to a Video CD, upload it to a Web site for a client presentation—the uses are only limited by the imagination.

Know Your Formats

Digital camcorders are all the rage now; the analogue version, while cheaper, doesn't offer the flexibility or the features that a digital one does. You will thus find the market saturated with digital models from all leading vendors.

Before investing in either of the available technologies it is important to learn which recording medium you go for. Ensure that the total cost of ownership of your purchase does not go through the roof because while you bought an affordable camcorder, the recording media costs a limb, and then some.

With that in mind, here are some of the common media available, their shortcomings and their features.

8mm

This format, about the size of an audio cassette, is used in analogue camcorders. The biggest advantage with this format is that the camcorders can be very compact. You can even carry a compact 8mm camcorder in your pocket—they can be that small and convenient. The format carries the same resolution as that of the stan-



This camera offers image stabilisation to shaky shots



Inside Stuff: How A Camcorder Works

